Functional Specification Document (FSD)

1. Introduction

Project Title: Automated Employee Onboarding & Offboarding System (ServiceNow)

Team Members:

- Kandula Uma Maheshwari (Lead Developer / ServiceNow Admin)
- HR / IT / Facilities / Security Stakeholders (Requirement Providers)

2. Project Overview

Purpose: The purpose of this project is to automate the manual, disjointed process of employee onboarding and offboarding. By leveraging ServiceNow, the system eliminates email-based coordination, ensures no tasks are missed, and provides a fully auditable lifecycle.

Features:

- Single self-service Service Portal intake form (13 variables).
- Dynamic UI Policies (showing Joining Date for Onboarding, Exit Date for Offboarding).
- Automated Manager Approval routing.
- Automatic creation of 3 departmental tasks (IT, Facilities, Security).
- Real-time status tracking via a custom Employee Lifecycle table.
- Role-Based Access Controls (ACLs) for data security.

3. Architecture

(Note: Adapted from React/Node to ServiceNow Low-Code)

- **Frontend (Presentation Layer):** Built using ServiceNow's Service Portal (AngularJS-based). It features a responsive, user-friendly catalog item form with dynamic UI policies. Agent-facing UI is used by Managers and Fulfillers.
- **Backend (Application Logic):** Built using ServiceNow Flow Designer (No custom scripts). It consists of a 15-action workflow triggered by the catalog submission, handling approvals, record creation, task generation, and status updates.
- **Database (Data Layer):** Built on ServiceNow's Cloud MySQL-based database. The primary custom table is u_employee_lifecycle. The system also interacts with Out-of-the-Box (OOB) tables like sc_request, sc_req_item, and sc_task.

4. Setup Instructions

Prerequisites:

- A ServiceNow Personal Developer Instance (PDI).
- Admin access to the instance.
- No external software (like Node.js or MongoDB) is required.

Installation (Replication Guide):

1. Log in to your ServiceNow PDI.
2. Create the custom table u_employee_lifecycle with required fields (Employee ID, Department, Manager, etc.).
3. Create Groups (HR, IT, Facilities, Security) and assign roles.
4. Build the "Employee Onboarding" Catalog Item with 13 variables.
5. Build the Flow Designer workflow (15 actions).

6. Configure the required ACLs under security_admin elevation.

5. Folder Structure

(This represents the GitHub Repository structure for the project)

- **/docs (Equivalent to Client/Server docs):** Contains all 19 markdown documentation files (Ideation, Design, Requirements, Testing, etc.).
- **/Screenshots:** Contains sub-folders (phase2, phase3, phase4, phase5) with all visual evidence of the system working.
- **/update-sets:** Contains the exported XML update set to easily migrate the application to another ServiceNow instance.
- **README.md:** The main project overview and links to all documentation.

6. Running the Application

(No terminal commands like npm start are required for ServiceNow)

To run the application: Log into your ServiceNow instance and navigate to the Service Portal.

- **Frontend URL (Service Portal):** https://dev396632.service-now.com/sp
- **Backend / Agent URL (Main Instance):** https://dev396632.service-now.com

7. API Documentation

(ServiceNow uses internal APIs and Flow Actions rather than traditional REST endpoints)

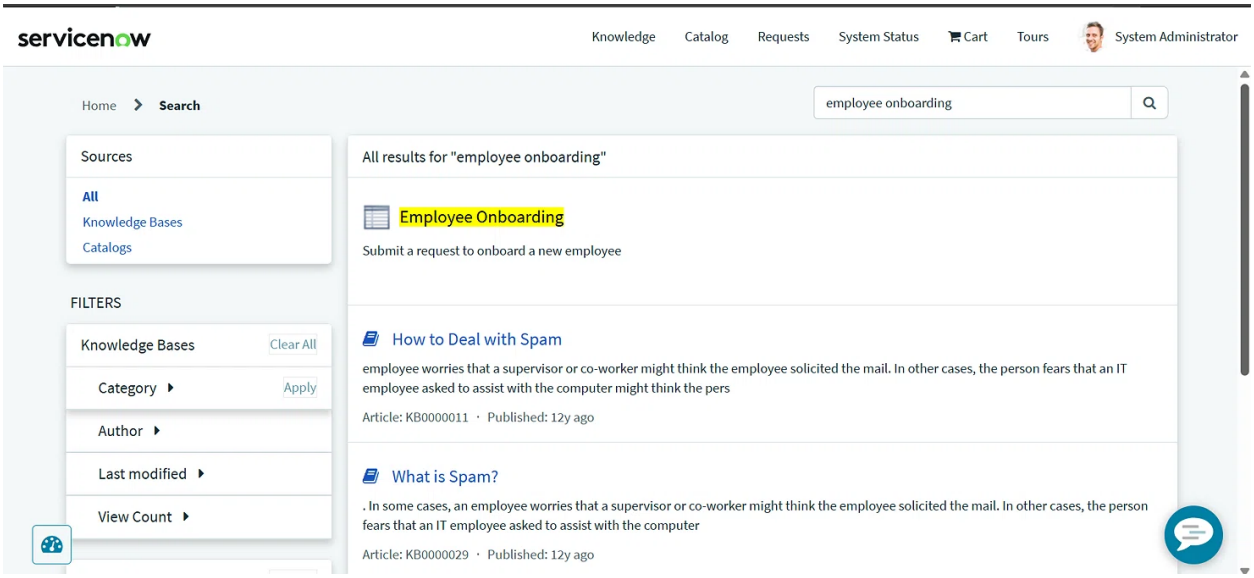
- **Integration:** The system uses ServiceNow's internal Table API to fetch data from u_employee_lifecycle.
- **Flow Actions:**
  - **Get Catalog Variables:** Retrieves the payload from the Service Catalog.
  - **Create Record:** Pushes data to the database.
  - **Ask For Approval:** Interfaces with the sysapproval_approver table.
- **(Future):** External REST APIs via Integration Hub will be used to connect to Active Directory.

8. Authentication

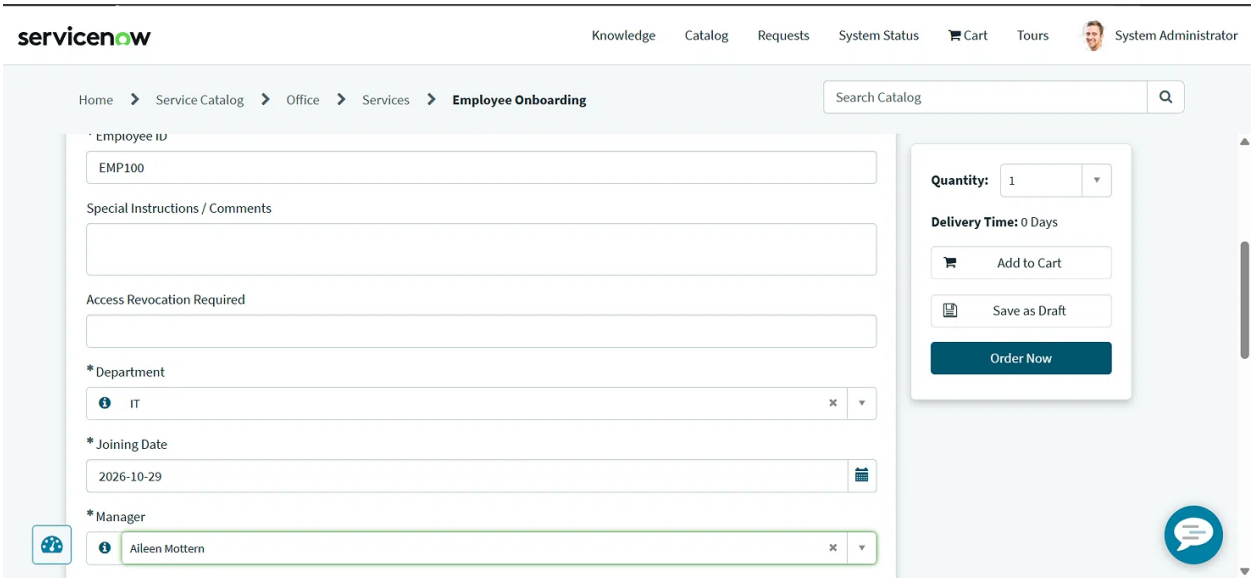
- **Authentication:** Handled entirely by ServiceNow's native security. Users log in via ServiceNow SSO or Local Login.
- **Authorization (ACLs):**
  - Access is governed by Access Control Lists (ACLs) applied to the u_employee_lifecycle table.
  - Read/Write/Create/Delete operations are restricted to users with the admin role.
  - Task visibility is restricted by Assignment Group membership (e.g., only the IT group can see IT tasks).

9. User Interface

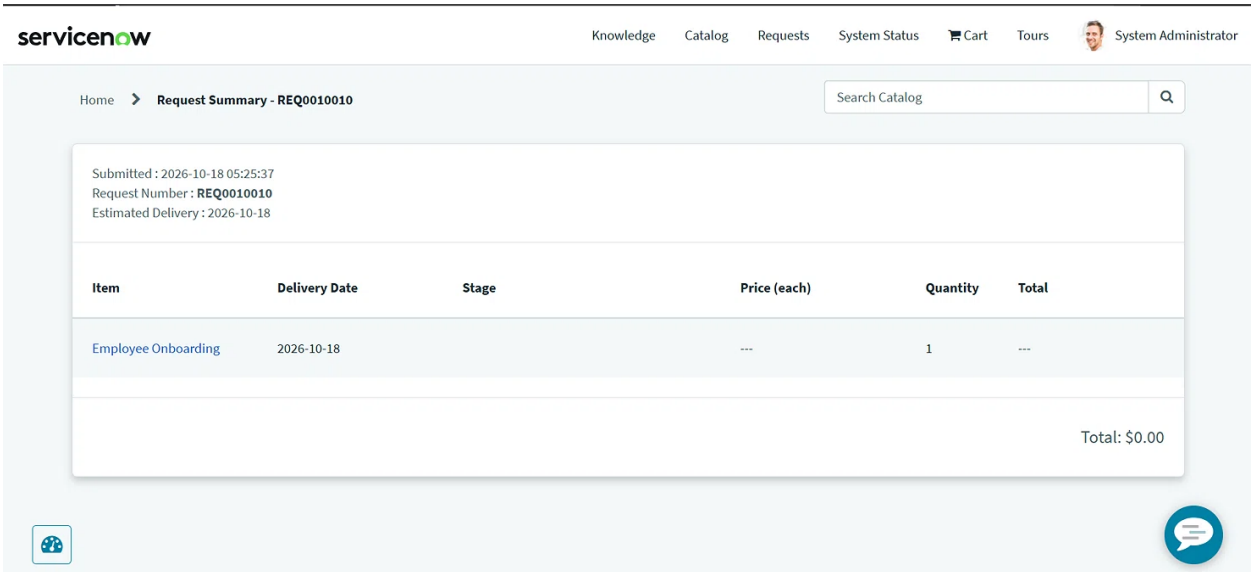
Screenshot 1: Service Portal Search.



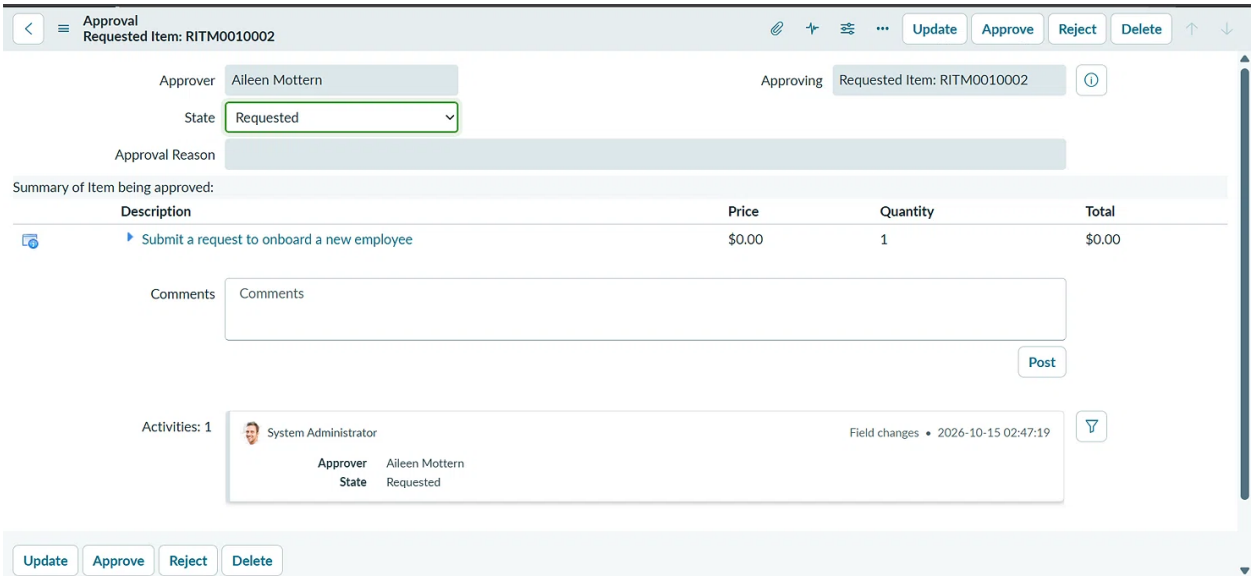
Screenshot 2: Dynamic Form (Onboarding selected).



Screenshot 3: Request Summary (REQ Number).



Screenshot 4: Manager Approval View.



10. Testing

- **Testing Strategy:** Unit Testing, Integration Testing, and User Acceptance Testing (UAT).
- **Tools Used:** ServiceNow Flow Execution Details (for backend debugging), UAT Test Cases (16 test cases passed), and Sys Email Logs.
- **Verification:** A full end-to-end test was completed successfully (REQ0010002 → RITM0010002 → Approved → Tasks Closed → Lifecycle Completed).

11. Screenshots or Demo

• **GitHub Repository:**
https://github.com/UmaMaheshwariKandula/servicenow-onboarding-offboarding

• **Demo Video:**
https://drive.google.com/file/d/15bFoSLjFWv8efrGt3p6yLLCJCoEJXH1O/view?usp=sharing

12. Known Issues

- **PDI Email Restriction:** The free Personal Developer Instance (PDI) has outbound email disabled by default. The flow's Send Email action executes perfectly, but actual emails are not delivered externally.
- **Single Record Processing:** The flow is designed for one onboarding request at a time. Bulk uploads are not natively supported in this version.

13. Future Enhancements

- **Active Directory Integration:** Use Integration Hub to auto-provision AD accounts.
- **Expanded HR Services:** Add Leave Requests and Transfer Requests using the same Flow Designer pattern.
- **Performance Analytics:** Build advanced dashboards for executive reporting on SLA compliance.
- **Virtual Agent:** Enable MS Teams/Slack chatbot integration for managers to approve requests without opening ServiceNow.